Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of a Conversational Agentic AI-based Sensor-Fusion Autonomous Optimization Intelligent Refrigerant Recovery System**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and Materials		*Information & Communications Technology (ICT)*	
	Medium Category	Small Category	Medium Category	Small Category
	Thermal & Fluid Process Equipment	*Refrigerant Recovery System*	*Artificial Intelligence*	*Agentic AI for Autonomous Process Optimization*
Concept of Proposed Technology	**Core Concept** This project aims to develop a conversational Agentic AI–based sensor-fusion, self-optimizing intelligent refrigerant recovery system, which integrates multi-sensor data in real time and enables Agentic AI to autonomously optimize the recovery process. Through this approach, the system simultaneously ensures high recovery quality and reliable traceability management, providing an integrated solution that overcomes the limitations of conventional manual recovery operations. Furthermore, all measurement and operational data generated during the recovery process are automatically recorded, validated, and transmitted, thereby establishing a digital management infrastructure that can be directly linked with national refrigerant management platforms such as the Refrigerant Information Management System (RIMS). This enables transparent, tamper-resistant tracking of refrigerant recovery history and supports institutional compliance with national greenhouse-gas reduction policies.			

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Main Functions

1) Real-Time Measurement and Multi-Sensor Fusion
   - Real-time acquisition of key physical parameters during refrigerant recovery, including pressure, temperature, weight, and sound signals
   - Time-synchronization and fusion of multi-sensor data to accurately recognize recovery status, refrigerant characteristics, and abnormal operating conditions
2) Agentic AI–Based Autonomous Operation and Optimization
   - AI-driven inference of refrigerant type, mixture condition, and potential contamination to determine optimal recovery strategies
   - Autonomous optimization of recovery speed, pressure control, and termination timing
   - Minimization of human errors and assurance of consistent recovery quality
3) Recovered Refrigerant Quality Recognition and Classification
   - On-site classification of recovered refrigerants into “Reusable / Reclamation Required / Disposal Required” through analysis of process data
   - Decision support for downstream purification and recycling processes
4) Automated Traceability Management and RIMS-Linked Reporting
   - Automatic recording of recovery amount, time, refrigerant characteristics, operator information, and equipment status
   - Tamper-resistant data management and automated electronic reporting
   - Direct compatibility with the national Refrigerant Information Management System (RIMS)
5) Conversational User Interface (Operator & Administrator Functions)
   - Voice- and text-based interaction between operators and the system
   - Automated provision of operation guidance, anomaly alerts, and recovery result reports
   - Enhanced usability, operator training efficiency, and on-site operational convenience

Potential Applications

This technology can be applied to refrigerant recovery operations in maintenance and disposal sites of refrigeration and air-conditioning systems. It enables data-based verification of refrigerant recovery performance and standardization of recovery quality, thereby supporting compliance with legal refrigerant recovery obligations.

In addition, by classifying the quality of recovered refrigerants, it enhances the efficiency of regeneration and purification processes and helps establish a stable supply system for reclaimed refrigerants. Furthermore, by building a management framework based on HFC reduction and refrigerant recovery performance data, the technology supports the implementation of national greenhouse gas reduction policies and provides a technical foundation for responding to international agreements and domestic regulations.

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Summary of Expected Effects

This technology can contribute to greenhouse gas reduction by minimizing refrigerant leakage and improving recovery rates. It also ensures stable quality of recovered refrigerants, increasing the proportion of refrigerants suitable for reuse and reclamation, while reducing dependence on operator skill and enhancing operational safety. Furthermore, it supports the advancement of data-based refrigerant management policies and is expected to strengthen domestic industrial competitiveness by creating intelligent recovery equipment and reclaimed refrigerant industries.

Figure. Overview of sensor-fusion autonomous optimization Intelligent refrigerant recovery system

Rationale for Joint R&D Project

Critical Technical Barriers

Fluorinated refrigerants (HFCs, etc.) account for a significant portion of national greenhouse gas emissions. In particular, emissions from the use of ODS substitute substances have recently increased sharply, making the advancement of refrigerant recovery and recycling technologies a key task for achieving national greenhouse gas reduction targets. However, the current refrigerant recovery market faces simultaneous technical and institutional limitations, including manually operated equipment structures, non-uniform recovery quality, limited reliability of traceability management, and insufficient linkage with the national management system (RIMS). These challenges are difficult to resolve through a single technology or a single institution alone, and therefore require convergent research and development integrating AI-based control technologies, sensor-fusion hardware technologies, refrigerant process domain expertise, and data integrity and security technologies.

Why Global Collaboration?

This project aims to develop a conversational Agentic AI–based sensor-fusion, self-optimizing

	intelligent refrigerant recovery system. To achieve this goal, the following multidisciplinary collaborations are required: 1) Refrigeration, air-conditioning, and refrigerant process specialist institutions: Recovery equipment design, refrigerant property analysis, and recovery process optimization 2) AI and Agentic AI research institutions: Multi-sensor data analysis, autonomous decision-making algorithms, and conversational interface development 3) ICT and data security specialist institutions: Real-time data acquisition and transmission, traceability management, and implementation of data integrity and tamper-prevention technologies 4) Policy and institutional linkage organizations: RIMS integration, verification of applicability to recovery management systems, and field demonstration The integrated development of industrial process–AI–ICT–policy linkage technologies involves high development risks and trial-and-error if conducted by a single institution. Therefore, joint research is essential to achieve technology convergence and to consolidate complementary capabilities. **Expected Impact** 1) Enhancement of technological completeness through the combination of advanced AI and autonomous control technologies with domestic refrigerant recovery systems and market environments 2) Expansion of early commercialization and industrial dissemination through demonstration-based technology development 3) Securing leading technologies for responding to global environmental regulations and establishing foundations for international standardization through joint research *4)* Acquisition of strategic core infrastructure technologies supporting national greenhouse gas reduction implementation
Key Words	1. Refrigerant Recovery System 2. Agentic AI 3. Sensor Fusion (Temperature / Pressure / Sound) 4. Autonomous Optimization 5. Conversational Control Interface

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)

R&D Objectives

R&D Objectives

The objective of this joint research is to develop a conversational Agentic AI–based sensor-fusion, self-optimizing intelligent refrigerant recovery system that autonomously controls and optimizes the refrigerant recovery process and automatically records, verifies, and transmits recovery history data. Through this, the project aims to establish a reliable recovery infrastructure capable of standardizing recovery quality and linking with the national Refrigerant Information Management System (RIMS).

Detailed Objectives

1) Development of multi-sensor–based real-time recovery state measurement and sensor-fusion technology
2) Development of Agentic AI–based recognition and autonomous operation control technology for identifying refrigerant type, contamination, and mixture conditions
3) Development of autonomous recovery process optimization algorithms and human-error minimization technologies
4) Development of automatic recovery amount and operation history recording and RIMS-linked data transmission technologies
5) Development of a conversational user-interface–based on-site operation support system
6) Fabrication and demonstration of an integrated intelligent refrigerant recovery system prototype

R&D Target Specifications

Category	Quantitative Target
Recovery efficiency (minimizing losses such as leakage)	≥ 95%
Refrigerant type recognition accuracy	≥ 98%
Mixture/contaminated refrigerant classification accuracy	≥ 95%
Recovery amount measurement error	≤ ±1%
Autonomous operation control success rate	≥ 95%
Automatic recovery history recording rate	100% (no manual recording required)
Conversational interface response accuracy	≥ 95%

R&D Plan

Year 1: Core Technology Development and Integrated System Design

1) Analysis of refrigerant recovery process requirements and system architecture design
2) Development of real-time measurement modules based on pressure, temperature, weight, and sound sensors
3) Development of sensor data synchronization and fusion algorithms
4) Development of Agentic AI–based refrigerant characteristic recognition
   - Recognition of refrigerant phase states based on temperature and pressure

conditions

- Design of recovery history data structure and RIMS-linked interface

5) Major Milestones
   - M1: Completion of multi-sensor measurement and data acquisition modules
   - M2: Laboratory validation of refrigerant recognition AI algorithms
   - M3: Completion of recovery history data automatic recording prototype

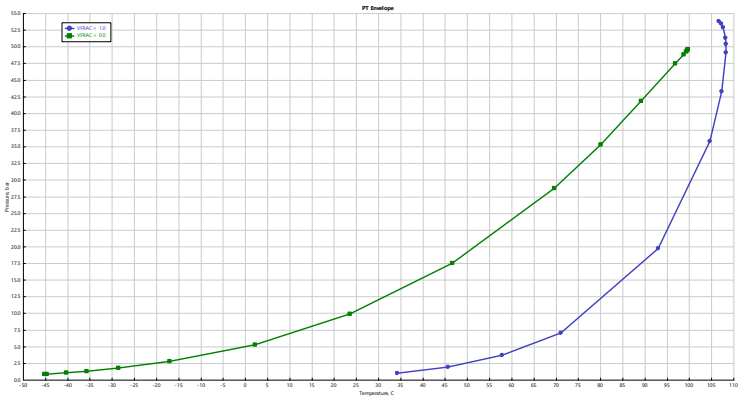

Figure. Graph of R-22 Waste Refrigerant Phase Behavior According to Temperature and Pressure

Year 2: Autonomous Optimization and Conversational Interface Development

1) Implementation of AI-based operation assistance functions to minimize human errors
2) Development of conversational (voice and text) interfaces for administrators to verify data integrity of acquired data
3) Application of data integrity, security, and tamper-detection technologies
4) Fabrication of integrated system prototype
5) Major Milestones
   - M4: Completion of Agentic AI–based operation control functions
   - M5: Completion of conversational interface
   - M6: Completion of integrated prototype fabrication

Year 3: Field Demonstration and Performance Verification

1) Field demonstration tests at industrial refrigeration and air-conditioning facilities and disposal sites
2) Verification of recovery efficiency, quality recognition, and traceability management performance
3) Demonstration of real-time data transmission linked with RIMS
4) System advancement reflecting user feedback
5) Establishment of commercialization design and standardization foundation
6) Major Milestones
   - M7: Completion of field demonstration and verification of target performance achievement
   - M8: Achievement of Technology Readiness Level (TRL) 6
   - M9: Establishment of commercialization model and business strategy

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institution develops Agentic AI–based autonomous optimization algorithms and sensor-fusion data processing technologies. It is also responsible for establishing intelligent strategies for the refrigerant recovery process, developing multi-sensor data analysis and refrigerant characteristic recognition models, and implementing the core engines for control logic and conversational interfaces. In addition, it designs the recovery history data structure, develops data integrity and tamper-prevention technologies, and RIMS-linked data transmission protocols, and leads the integrated system design, establishment of performance verification indicators, and performance evaluation in laboratory and pilot environments. The research institution also undertakes the establishment of the joint research technology roadmap, algorithm advancement, performance verification, technical standardization, and creation of academic and patent outcomes, and enhances technological completeness through regular technical consultations and integrated testing with Korean SME partners.
Expected R&R of Korean SME Partner	The Korean SME partner leads the development and demonstration of field-applicable hardware systems for this project, and is responsible for the mechanical design and fabrication of refrigerant recovery equipment, construction of pressure, temperature, weight, and sound sensor modules, and development of recovery controllers and ICT communication modules. In addition, the partner applies the prototype system to actual refrigeration and air-conditioning maintenance and disposal sites to verify recovery performance, safety, operational convenience, and durability, and provides recovery process operation data and demonstration results to the research institution to support AI algorithm training and advancement. The Korean SME partner also carries out field demonstration of the automatic recovery history recording function and RIMS-linked system, collection of user feedback, improvement of productization design, and establishment of commercialization and market entry strategies, and undertakes business development and certification activities for the expansion of the domestic refrigerant recovery industry.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The conversational Agentic AI–based sensor-fusion, self-optimizing intelligent refrigerant recovery system developed through this joint research is expected to generate strategic core outcomes that go beyond technological achievements, simultaneously advancing domestic industrial sophistication and enabling entry into the global eco-friendly market. **Technological Achievements** 1) Securing core technologies for real-time recovery state recognition and automatic refrigerant characteristic identification based on multi-sensor fusion 2) Development of Agentic AI–based optimized recovery control algorithms 3) Securing automatic recovery history recording and data integrity and tamper-prevention technologies 4) Implementation of a digital recovery management infrastructure compatible with the national Refrigerant Information Management System (RIMS) and international refrigerant management systems 5) Fabrication and field demonstration of an integrated intelligent refrigerant recovery system prototype (achievement of TRL6) **Commercialization Potential** 1) Expansion of intelligent recovery equipment supply to domestic refrigerant recovery companies and maintenance and disposal markets 2) Activation of the reclaimed refrigerant industry and creation of new service markets through securing recovered refrigerant quality 3) Development of package-type business models combining recovery equipment, data management, and AI software 4) Expansion of subscription-based businesses for maintenance services, data management platforms, and AI operation software **Global Industrial Expansion** 1) Rapid growth expected in the global refrigerant recovery and recycling market due to strengthened HFC reduction regulations under the Kigali Amendment to the Montreal Protocol 2) The technology includes AI autonomous control, data integrity, and traceability management functions, enabling simultaneous compliance with international environmental regulations and certification requirements 3) Establishment of a foundation for entering refrigerant recovery and reclamation markets in advanced countries such as North America, Europe, and Japan 4) Supply of export-oriented intelligent recovery equipment to emerging markets in Southeast Asia, the Middle East, and South America, where refrigeration and air-conditioning facilities are rapidly increasing 5) Potential expansion into international standardization (ISO/IEC), global carbon reduction

	certification, and carbon credit–linked businesses in the future **Environmental and Policy Impact** 1) Contribution to achieving national greenhouse gas reduction targets by minimizing HFC leakage and improving recovery rates 2) Support for advancement of domestic refrigerant management systems through securing reliability of recovery performance data 3) Provision of a technological foundation for implementing international environmental agreements and participation in global carbon reduction cooperation **Industrial and Economic Impact** 1) Promotion of intelligent recovery equipment, AI, and ICT convergence industries and strengthening technological competitiveness of domestic SMEs 2) Facilitation of a reclaimed refrigerant circular economy and creation of high value-added green industries 3) Securing early entry into global eco-friendly equipment markets and expansion of export industries 4) Training of specialized personnel and creation of new industrial jobs

6. References

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